James Utley
EC500 - Computer Vision

Homework 3 - Writeup

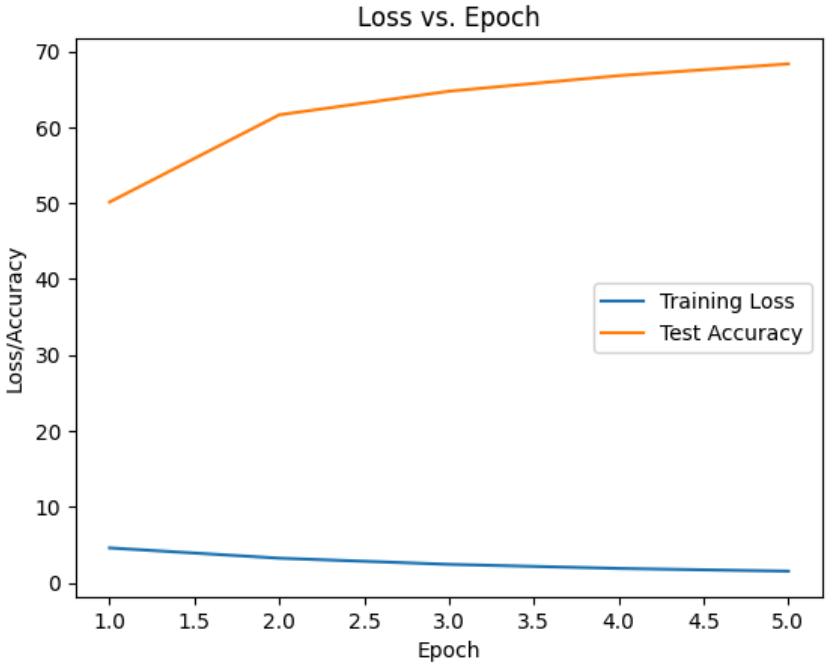
Part 1

Job 4

Description of Observations (150 words max)

The plot below shows the loss and accuracy by epoch for the linear probe of ResNet50. Training loss decreases monotonically and test accuracy increases monotonically with each epoch. ResNet50 achieves around 67% test accuracy after five epochs. Both the training loss and the test accuracy approach respective values asymptotically, approximately 0 for training loss and 70 for test accuracy. If training were continued to more epochs, the loss and accuracy would likely continue to approach these values, until the model begins to overfit at which point the test accuracy will begin to decrease.

Plots



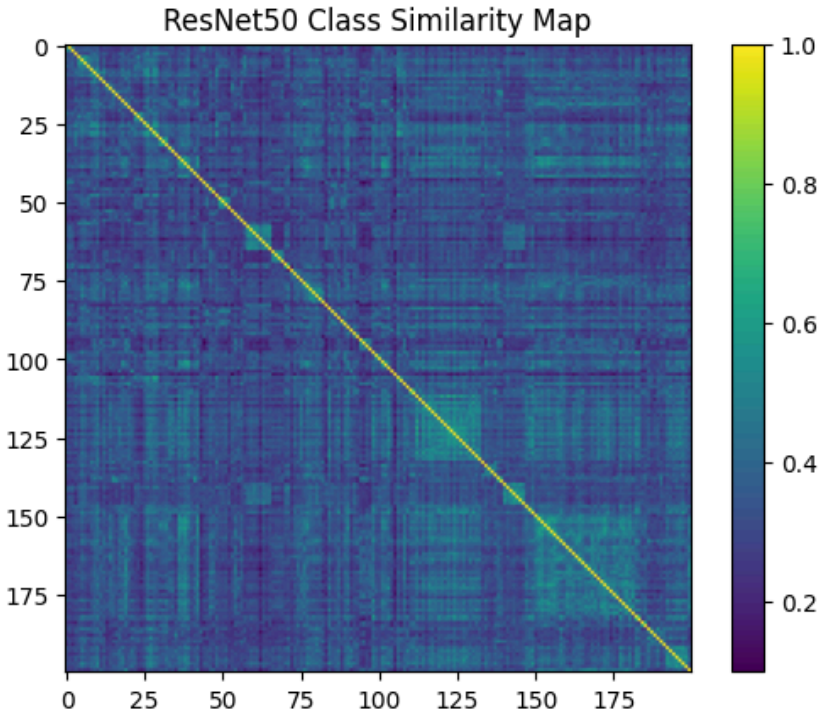
Job 5

Description of Observations (150 words max)

The plot below shows the cosine similarity between each element in the learned vector W. Along the diagonal, the cosine similarity is 1, since these vectors are the same by definition. The higher the value, the more similar each element of that comparison of W is. There are regions of rectilinear similarity where it appears that vectors in that square are more likely to be similar, such as the green squares that appear near the diagonal.

As an aside, this cosine similarity graph reminds me of a very intricate plaid design.

Plots



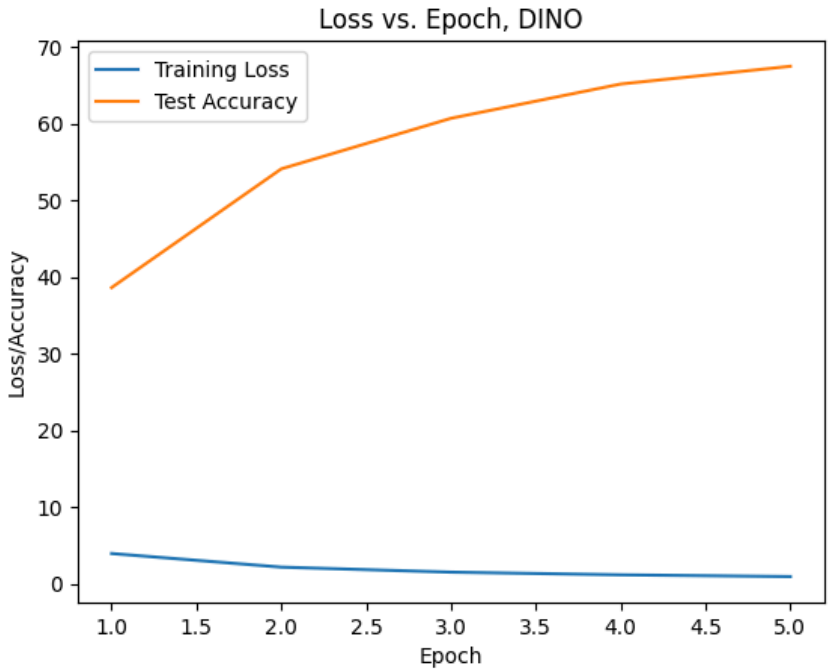
Job 6

Description of Observations of Job 4 implementation in DINOv2 (150 words max)

The below graph shows training loss and test accuracy for the linear probe implementation on DINOv2. The test accuracy in epoch one is lower than the corresponding accuracy for ResNet50, but, after epoch five, DINOv2's test accuracy was about 68%, which is comparable to the test accuracy of ResNet50. As in the graph for ResNet50, the test accuracy increases monotonically and the training loss decreases monotonically. Both the training loss and the test accuracy appear to be asymptotically approaching respective values, zero for the loss and perhaps 70 for the accuracy. Should we train for more epochs, we

would see these values further approach their asymptotic values, until the model begins to overfit, which would then lead to a decrease in test accuracy.

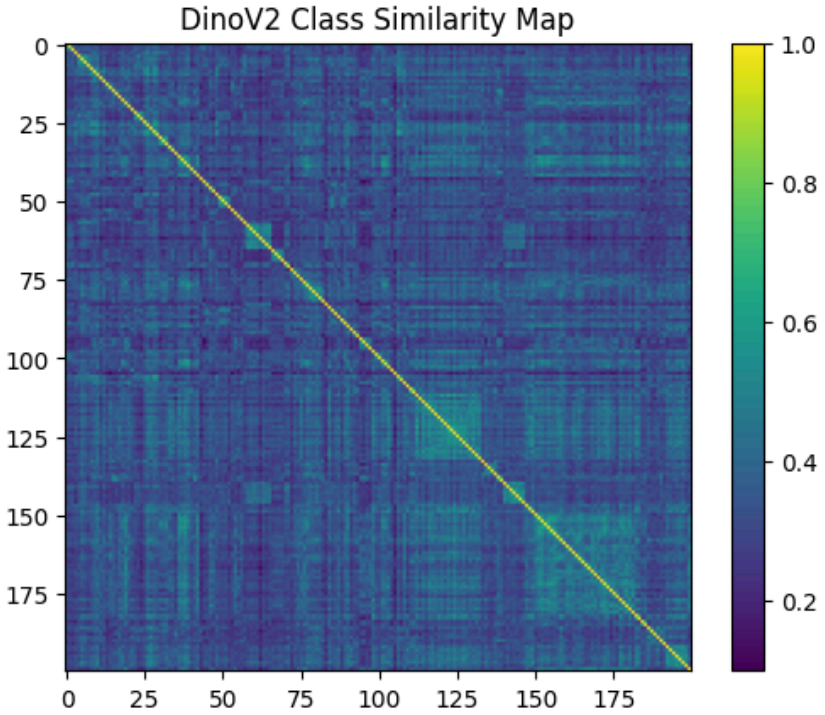
Plots



Description of Observations of Job 5 implementation in DINOv2 (150 words max)

The graph below shows the cosine similarity between each element of the learned vector W in the linear probe of DINOv2. Like for the cosine similarity graph for ResNet50, the values along the diagonal equal one because they are comparing the same vector, and there exist rectilinear patches where the similarity value is higher. In fact, these two graphs looked so similar to the naked eye that I checked three times to make sure I was plotting the correct values. The similarity between the ResNet50 and DINOv2 similarity maps, as well as the models' similar final test accuracies, indicates to me that these two linear probes learn nearly identical vectors W.

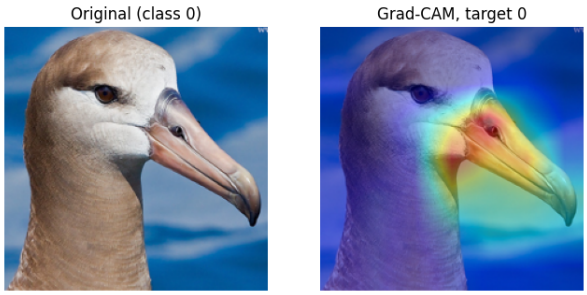
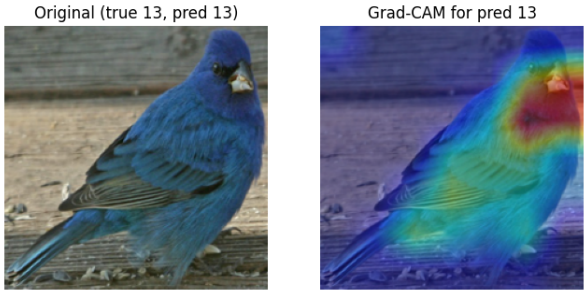
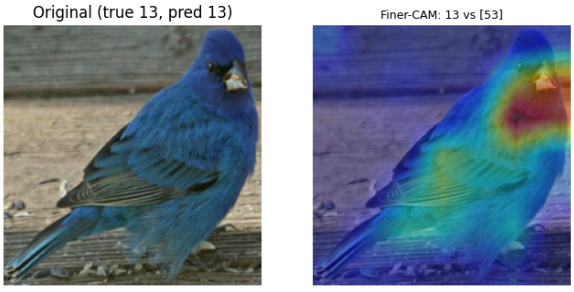
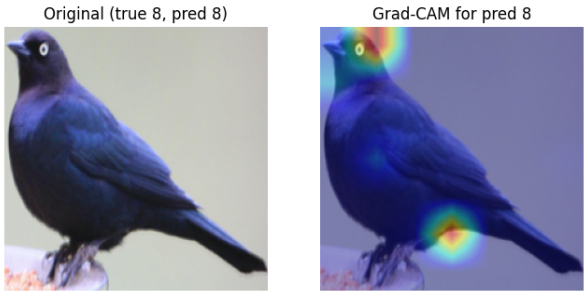
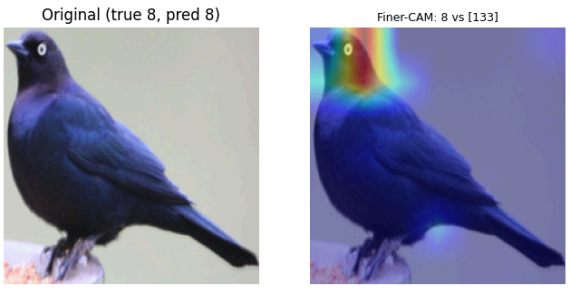
Plots



Part 2

Since part 2 deals with the comparison between Grad-CAM and Finer-CAM, I have structured this passage differently so that the comparisons are easier to see.

Job 7	**Job 8**
Description of Observation (250 words max)	Description of Observation (250 words max)
Grad-CAM successfully isolates feature vectors that identify aspects of the class they are meant to identify. However, as we discussed in class, the model is not as good at picking out specific features endemic to a particular class. This can be seen in the heat maps below which show the locations of the object in the image. The heat maps are generally accurate to the object, but they are not precise about what features define that class in the object. This can lead to misclassification of birds during testing and drive down the accuracy of the model.	In contrast, Finer-CAM isolates class features present in the object. Like Grad-CAM, Finer-CAM correctly identifies the objects in the image. However, unlike Grad-CAM, Finer-CAM can accurately make comparisons between different classes and generate feature maps of the presence of distinctive characteristics for a particular class (seen below as heat maps overlaid on the original image). These feature maps are more precise than the ones present in Grad-CAM, and importantly they can identify features that differentiate classes. This can be seen below in the

This definition presents itself in the images below. The heat maps identify the bird in each image, but the presentation of the heat map is only for showing the particular class. Notice the label above the heat map for Grad-CAM versus those for Finer-CAM: Grad-CAM only identifies one class in the label whereas Finer-CAM compares the correct class versus an incorrect one. This shows that Grad-CAM cannot make comparisons between the classes.	presentation of the heat maps where the heat map identifies differences between classes. This precision allows Finer-CAM the ability to specify and differentiate between species and allows for a more generalizable model.
Plots	Plots
	
	
	

AI Usage Disclosure (150 words max)

I used Claude Opus 5.5 (high effort) in the following ways. Transcripts for the chats can be provided on request.

- I used Claude to understand crossEntropyLoss for criterion (Part 1, Job 1)
- I used Claude to understand cosine similarity (I totally knew what it was, I just forgot in the moment and panicked…) (Part 1, Job 5)
- After drafting my best attempt, I ran def grad_cam() through claude to have it help me use the correct syntax for target, but I had it guide me through the corrections without providing the code rather than just providing me a rewrite of the code. (Part 2, Job 7)
- I used Claude to understand the meaning behind the computation for Sub section 4 in part 2 Job 8
- I used Claude to point out errors and bugs in my version of sub part 3 in Part 2 Job 8

Student Collaborators

None. I completed this on my own.

I did send this homework to a friend, James Wareham, who was interested in Grad-CAM vs. Finer-CAM, but since he is not in the course, I do not believe that it should be a problem.